

Zach Prynne

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EDUCATION

University of Manitoba

Winnipeg, MB

Bachelor of Science in Computer Science, Minor in Mathematics

Jan. 2022 – Present

- **Coursework:** Data Structures, Algorithms, Computer Systems, Software Development, Programming Paradigms

EXPERIENCE

Explore Science Day Volunteer

Nov. 2025

University of Manitoba

Winnipeg, MB

- Facilitated hands-on workshops for high school students using the BBC Micro-bit microcontrollers to introduce fundamental concepts of Embedded Systems and physical computing
- Guided participants in interfacing software with hardware components, using onboard accelerometers, LED arrays, and sensors to build interactive, event-driven programs
- Explored and implemented creative pedagogical methods to visualize programming logic, effectively engaging diverse groups in introductory STEM topics

Keyholder

Aug. 2023 – Present

Buggy Bath

Winnipeg, MB

- Entrusted with facility keys to manage opening and closing procedures, including alarm system activation and secure handling of daily cash deposits
- Supervised high-volume daily operations, ensuring equipment functionality and maintaining service quality for over 200 vehicles per shift

Courier

Aug. 2019 – Aug. 2023

FedEx Express

Winnipeg, MB

- Managed time-sensitive logistics for 80+ daily deliveries, consistently meeting strict deadlines through efficient route optimization and priority management
- Utilized handheld scanning technology to maintain 100% data accuracy for real-time package tracking
- Resolved on-the-road logistical challenges and equipment troubleshooting, ensuring zero service interruptions

PROJECTS

Farm-Clicker | TypeScript, PGLite, Node, Git

Jan. 2026 – Present

- Developed a full-stack incremental game utilizing TypeScript
- Architected the application using MVC to decouple game logic from the UI, improving maintainability and scalability
- Constructed a multi-user authentication system using PGLite, implementing relational data to ensure unique game states and progress persistence between accounts
- Built a dynamic web interface with HTML5 and CSS, utilizing DOM manipulation to provide real-time visual feedback on production rates and currency accumulation.

Library Management System | Java, Maven, Git

Sept. 2025 – Dec. 2025

- Developed a modular Library Management System in Java, utilizing SOLID and DRY principles
- Designed a robust Domain Model to represent core model objects, ensuring clear separation of concerns and data integrity within the application
- Mapped Flows of Interaction to handle multi-step processes such as book checkouts, membership validation, and overdue book processing
- Implemented custom exception handling to manage invalid user inputs and edge cases, ensuring application stability during runtime
- Streamlined the development process using Maven for dependency management and Git for version control

TECHNICAL SKILLS

Languages: Java, C/C++, SQL (PGLite), JavaScript, HTML/CSS, R, MIPS

Frameworks: Node.js, PGLite, JUnit, Maven

Developer Tools: Git, VS Code, Visual Studio, IntelliJ, NeoVim